Two versions of the derivation

AC

March 26, 2026

1 Short Version (Better for PRL?)

Closed form for $d = 2$. For $d \geq 2$, by using the dual Cauchy Identity we can obtain a similar closed form in terms of distribution polynomial of a Mahonian Statistic - $\mathrm{maj}(\pi) - \mathrm{inv}(\pi), \pi \in S_n$. We first split the X into a product $X = \hat{X}\hat{X}$, where $\hat{X} = z_1^{N,1}$, giving us $\Xi_2^N(t) = \lambda_t(\hat{X}\hat{X})$. We can then relate $\lambda_t(\hat{X}\hat{X})$ to the dual Cauchy Identity [1]

$$\lambda_t(\hat{X}\hat{X}) = \sum_{\lambda} s_\lambda(b) s_{\lambda'}(b) t^{|\lambda|}$$

where s_λ are Schur functions and λ' denotes the conjugate partition. We can then write $Z_n^2(b)$ in terms of Schur functions by equating the powers of t.

$$Z_n^{N,2}(b) = \sum_{\lambda \vdash n} s_\lambda(b) s_{\lambda'}(b)$$

Using the SYT expansion of Schur functions [2, Cor. 7.21.5] via the hook-content formula and $\mathrm{maj}(T^t) = \binom{n}{2} - \mathrm{maj}(T)$, this becomes a sum over tableau pairs. Under the RSK correspondence [2, Eq. (7.23.1)], this maps to permutations $\pi \in S_n$ with statistics $\mathrm{maj}(\pi)$ and $\mathrm{maj}(\pi^{-1})$. Applying Theorem 1 of Ref. [3] then yields

$$Z_n^{N,2}(b) = \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\pi \in S_n} b^{\mathrm{maj}(\pi) - \mathrm{inv}(\pi)}. \tag{1}$$

2 Detail Version (How I wrote it before)

For $d \geq 2$, by using the dual Cauchy Identity we can obtain a similar closed form in terms of distribution polynomial of a Mahonian Statistic - $\mathrm{maj}(\pi) - \mathrm{inv}(\pi), \pi \in S_n$. We first split the X into a product $X = \hat{X}\hat{X}$, where $\hat{X} = z_1^{N,1}$, giving us $\Xi_2^N(t) = \lambda_t(\hat{X}\hat{X})$. We can then relate $\lambda_t(\hat{X}\hat{X})$ to the dual Cauchy Identity in terms of the 'Schur Opeartions' [1] and use it's algebraic variant in terms of Schur functions.

$$\lambda_t(\hat{X}\hat{X}) = \sum_{\lambda} S^\lambda(\hat{X}) S^{\lambda'}(\hat{X}) t^{|\lambda|} \xrightarrow{\text{Algebraic Variant}} \sum_{\lambda} s_\lambda(b) s_{\lambda'}(b) t^{|\lambda|}$$

where s_λ are Schur functions and λ' denotes the conjugate partition. We can then write $Z_n^2(b)$ in terms of Schur functions by equating the powers of t.

$$Z_n^{N,2}(b) = \sum_{|\lambda| = n} s_\lambda(b) s_{\lambda'}(b) = \sum_{\lambda \vdash n} s_\lambda(b) s_{\lambda'}(b)$$

Writing s_λ in terms of 'hook polynomial', $s_\lambda = b^{n(\lambda)} \prod_{u \in \lambda} (1 - b^{h(u)})^{-1}$ [1], and using corollary 7.21.5 from [2] we have

$$s_\lambda(b) = \frac{1}{(b)_n} \sum_{T \in \mathrm{SYT}(\lambda)} b^{\mathrm{maj}(T)}$$

For $s_{\lambda'}$ we note that there is a bijection between $T \in \mathrm{SYT}(\lambda)$ and $T^t \in \mathrm{SYT}(\lambda')$, hence we can get

$$
\begin{aligned}
s_{\lambda'}(b) &= \frac{1}{(b)_n} \sum_{T \in \mathrm{SYT}(\lambda)} b^{\mathrm{maj}(T^t)} \\
&= \frac{1}{(b)_n} \sum_{T \in \mathrm{SYT}(\lambda)} b^{\binom{n}{2} - \mathrm{maj}(T)}
\end{aligned}
$$

Here we used the fact, for $i \in [n-1]$, $i \in \mathrm{des}(T^t)$ if and only if $i \notin \mathrm{des}(T)$ [4], hence

$$
\begin{aligned}
\mathrm{maj}(T^t) &= \sum_{i \in \mathrm{des}(T^t)} i = \sum_{i \in [n-1] \setminus \mathrm{des}(T)} i \\
&= \sum_{i=1}^{n-1} i - \sum_{i \in \mathrm{des}(T)} i = \binom{n}{2} - \mathrm{maj}(T)
\end{aligned}
$$

combining the two, we get

$$
Z_n^{N,2}(b) = \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\lambda \vdash n} \sum_{(P,Q) \in SYT(\lambda)} b^{\mathrm{maj}(P) - \mathrm{maj}(Q)}
$$

Since both P and Q have the same shape we associate it with a permutation $\pi \in S_n$, under RSK (7.23.1 from [2]) we have $\mathrm{maj}(P) = \sum_{i \in \mathrm{des}(P)} i = \sum_{i \in \mathrm{des}(\pi^{-1})} i = \mathrm{maj}(\pi^{-1})$, (similarly $\mathrm{maj}(Q) = \mathrm{maj}(\pi)$). This gives

$$
\begin{aligned}
Z_n^{N,2}(b) &= \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\pi \in S_n} b^{\mathrm{maj}(\pi^{-1}) - \mathrm{maj}(\pi)} \\
&= \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\pi \in S_n} b^{\mathrm{maj}(\pi) - \mathrm{maj}(\pi^{-1})}
\end{aligned} \tag{2}
$$

The equality holds between the inverses switch since the sum contains all permutations and each permutation is an inverse of some permutation. This also means our resulting polynomials have to be palindromic.

Finally using Theorem 1 from [3] we can exchange $\mathrm{maj}(\pi)$ with $\mathrm{inv}(\varphi(\pi))$ while preserving $\mathrm{des}(\pi^{-1})$ under the mapping and hence $\mathrm{maj}(\pi^{-1})$. Then since we are summing over all elements of S_n we replace π with π^{-1} and using $\mathrm{inv}(\pi^{-1}) = \mathrm{inv}(\pi)$ [3] we arrive at the final distribution polynomial $F_n(b)$,

$$
\begin{aligned}
Z_n^{N,2}(b) &= \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\pi \in S_n} b^{\mathrm{inv}(\pi) - \mathrm{maj}(\pi)} \\
&= \frac{b^{\binom{n}{2}}}{((b)_n)^2} \sum_{\pi \in S_n} b^{\mathrm{maj}(\pi) - \mathrm{inv}(\pi)} = \frac{b^{\binom{n}{2}}}{((b)_n)^2} F_n(b)
\end{aligned} \tag{3}
$$

Again, bring back the zero point energy then gives $Z_n^{N,2} = b^{\binom{n+1}{2}} ((b)_n)^{-2} F_n(b)$.

References

[1] I.G. Macdonald. *Symmetric Functions and Hall Polynomials*. Oxford classic texts in the physical sciences. Clarendon Press, 1998.

[2] Richard P. Stanley and Sergey Fomin. *Enumerative Combinatorics*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 1999.

[3] Dominique Foata and Marcel Paul Schützenberger. Major index and inversion number of permutations. *Mathematische Nachrichten*, 83:143–159, 1978.

[4] Brice Huang. Cyclic descents for general skew tableaux. *Journal of Combinatorial Theory, Series A*, 169:105120, January 2020.